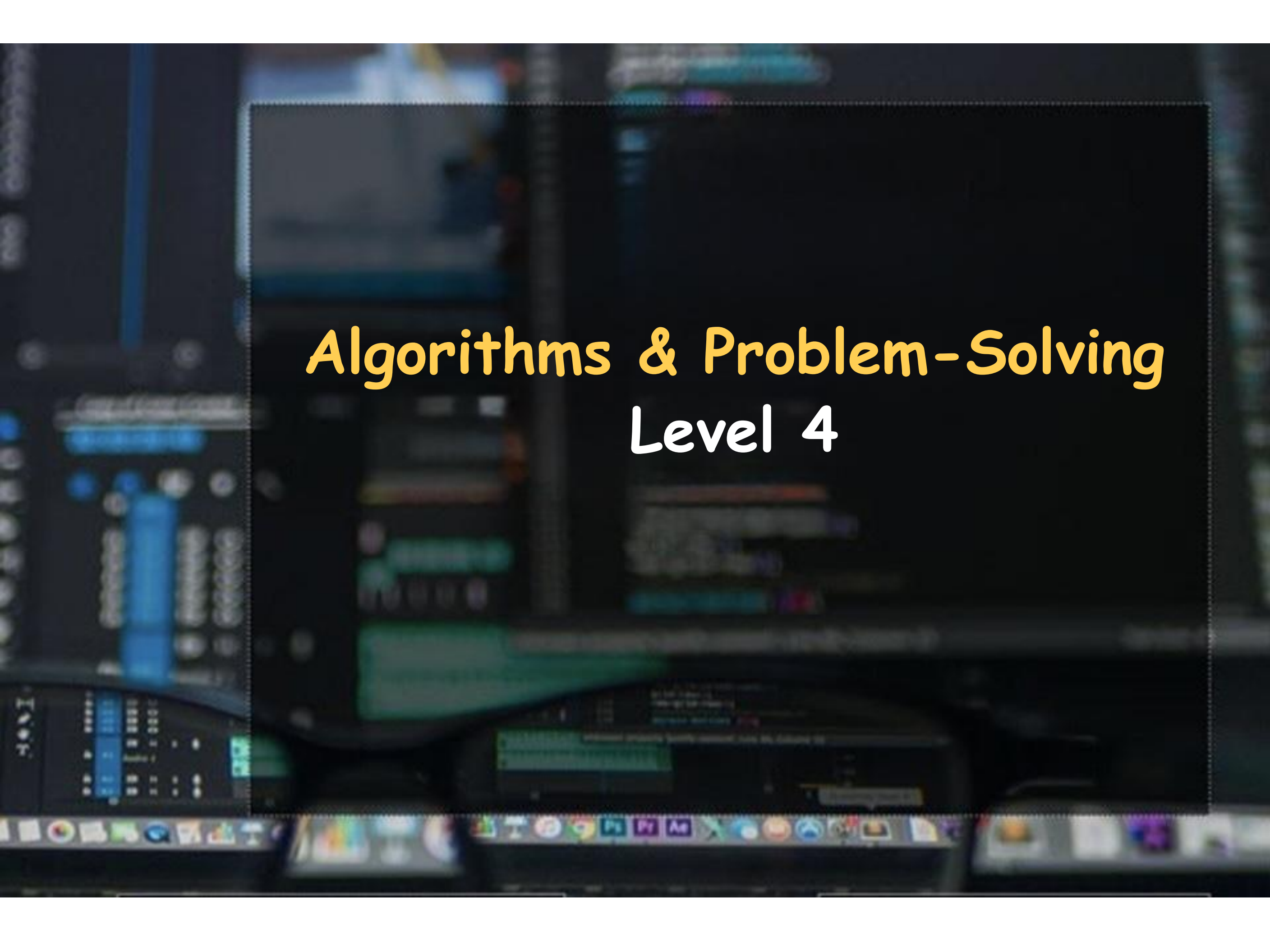
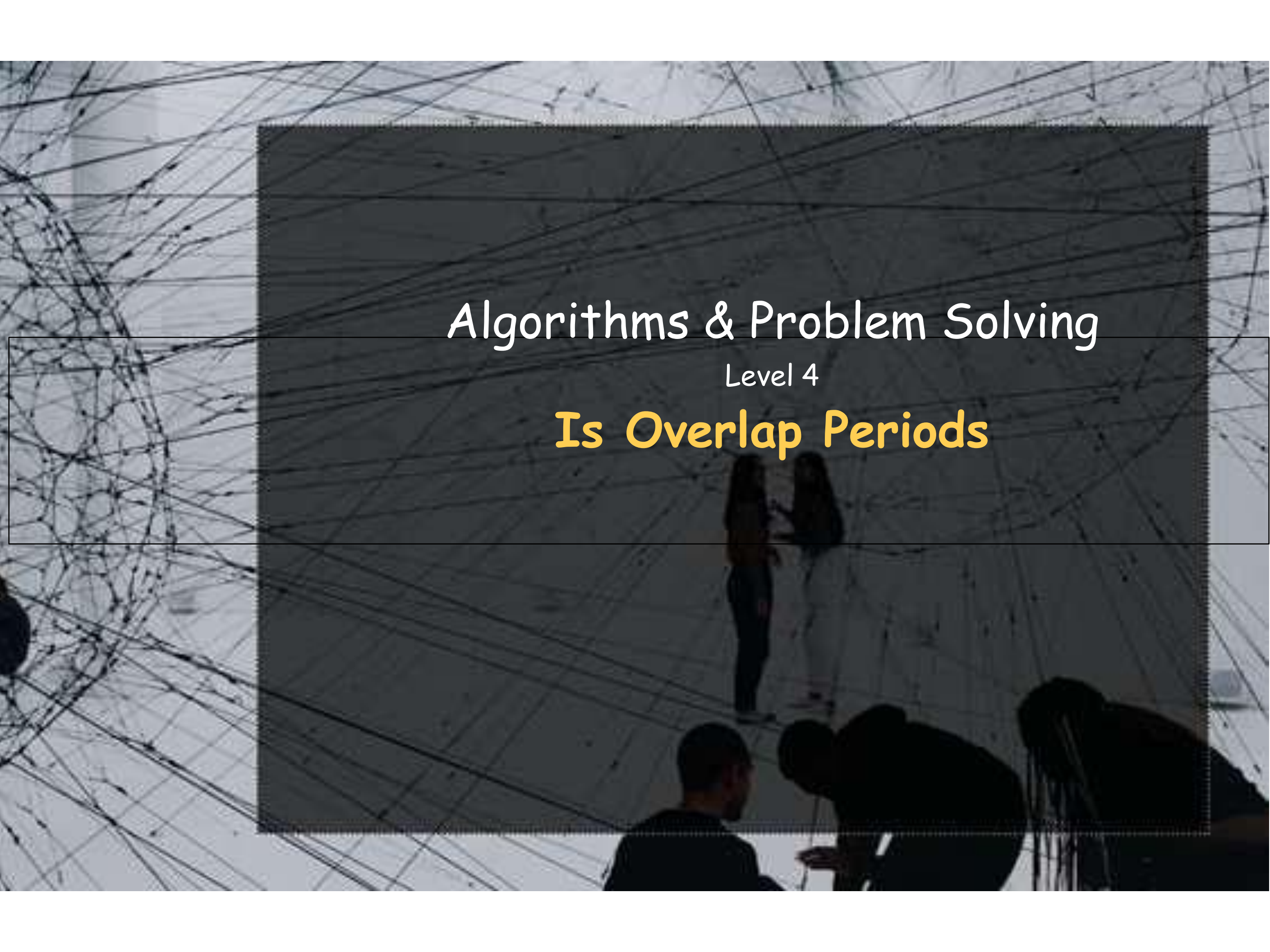




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, the dark silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, with some individuals leaning forward as if working on a table.

Algorithms & Problem Solving

Level 4

Is Overlap Periods

Definition

- sDate Struct: A structure containing the day, month, and year.
- sPeriod Struct: A structure containing two dates: ****StartDate**** and ****EndDate**** to represent a time period.
- enDateCompare Enum: An enumeration type that determines the result of comparing two dates:
 - Before = -1 (date before date)
 - Equal = 0 (dates equal)
 - After = 1 (date after date)



Explanation:

- This program reads two time periods (Period1 and Period2) and checks if the two periods overlap (i.e., if they share a common time interval).
- `CompareDates(date1, date2)`: Returns if the first date is before, after, or equal to the second.
- `IsOverlapPeriods(period1, period2)`: Checks for overlap between the two periods according to the following logic:
 - `CompareDates`: If the end of the second period is before the start of the first, there is no overlap.
 - `CompareDates`: If the start of the second period is after the end of the first, there is no overlap.
 - Otherwise, there is an overlap.

Procedures

- 1 Read the start and end dates of the first period
- 2 Read the start and end dates of the second period
- 3 Call the `IsOverlapPeriods` function to check for overlap
- 4 Print the appropriate result

Example:

Period 1: 1/2/2022 10/2/2022

Period 2: 5/2/2022 15/2/2022

Yes, Periods Overlap.

Explanation:

- Because the second period begins within the scope of the first period (between February 1st and 10th), there is a clear overlap.

Note:

 If one period ends exactly before the other begins, the result will be "No Overlap"

Summary

- ✅ The program determines if there is a time overlap between two periods using logical date comparison.
- ✅ It relies on comparing the start and end dates of each period. ✅ It does not depend on pre-built libraries but on logical time calculations based on manually entered values.

Thank you 😊